

Scientific Communities in the Literature Selection

Inductive thematic coding based on the uploaded literature table. Only the columns **title** and **keywords** were used.

Coding note

The dataset contains **250** records. The categories below are not intended as mutually exclusive article-level codes. Instead, they describe broader scientific communities that are visible in the literature selection. Individual articles may belong to more than one community. The optional counts are rough, non-exclusive orientation values based on keyword and title matching and should not be read as final coded frequencies.

Category system

No.	Scientific community	Definition	Example title	Approx. n
1	Educational Technology, AI & Learning Analytics	Research communities concerned with digital learning environments, artificial intelligence, machine learning, natural language processing, learning analytics, virtual or augmented reality, and data-driven educational technologies.	<i>Are U.S. graduate curricula ready to prepare social data scientists for the AI era?</i>	102
2	STEM, Mathematics & Science Education	Research on teaching and learning in mathematics, science, technology, engineering, computer science, bioinformatics, and STEM identity or participation.	<i>Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database</i>	78
3	Higher Education, College & Graduate Education	Studies of higher education, college students, doctoral and graduate education, academic careers, student retention, transitions into college, and career outcomes.	<i>Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education</i>	53
4	Assessment, Psychometrics & Educational Measurement	Research focused on tests, scales, diagnostics, screening, validation, psychometric modelling, measurement invariance, large-scale assessments, and student evaluations.	<i>Development and validation of a rapid and precise online sentence reading efficiency assessment</i>	52
5	Motivation, Self-Regulation & Student Well-Being	Work addressing motivational beliefs, self-efficacy, expectancy-value beliefs, emotion, stress, mental health, well-being, self-regulation, and related psychological learning conditions.	<i>The limited role of expectancy-value beliefs, self-efficacy, and perceived attentional control in predicting online learning outcomes in a general education course</i>	37
6	Teacher Education, Professional Development & Teacher Learning	Research on teacher education, professional development, teacher beliefs, teacher collaboration, professional learning networks, teacher agency, and instructional practice.	<i>A Different Experience in a Different Moment? Teachers Social Media Use Before and During the COVID-19 Pandemic</i>	41
7	Inclusion, Special Education, Disability & Equity	Communities concerned with inclusive education, special education, disability, diversity, social inequality, race, gender, first-generation students, gifted education, and educational equity.	<i>Effect of Local Norms on Racial and Ethnic Representation in Gifted Education</i>	35
8	COVID-19, Online Learning & Emergency Remote Education	Studies that explicitly examine pandemic conditions, lockdowns, online teaching, emergency remote education, stress, and students' or teachers' experiences during COVID-19.	<i>Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic</i>	18

No.	Scientific community	Definition	Example title	Approx. n
9	Curriculum, Textbooks, Discourse & Educational Content Analysis	Research analysing educational content, curricula, textbooks, public discourse, social media communication, reform debates, and representations in learning materials.	<i>Content Analysis of Textbooks via Natural Language Processing: Findings on Gender, Race, and Ethnicity in Texas U.S. History Textbooks</i>	13
10	Open Science, Research Methods & Evidence Synthesis	Methodologically oriented work on open science, preregistration, replication, systematic reviews, bibliometrics, meta-analyses, research quality, and methodological classification.	<i>Open Education Science</i>	30
11	Intervention, Program Evaluation & Experimental Education Research	Studies evaluating educational programs, reforms, interventions, or learning environments, often using experimental, quasi-experimental, field-experimental, or evaluation designs.	<i>An Evaluation of the Educational Impact of College Campus Visits: A Randomized Experiment</i>	45
12	School Policy, Reform & Educational Systems	Research addressing educational policy, reform, school choice, administrative data, systemic governance, school organisation, public sentiment, and structural conditions of education systems.	<i>Understanding Public Sentiment About Educational Reforms: The Next Generation Science Standards on Twitter</i>	9

Interpretive summary

Overall, the literature selection is strongly shaped by methodologically and data-oriented educational research. Educational technology, AI, learning analytics, assessment, psychometrics, open science, and evidence synthesis are highly visible. Content-wise, the selection also contains substantial communities around higher education, STEM education, motivation and well-being, teacher learning, inclusion, disability, and equity.

Use of the categories

For a subsequent formal coding process, these categories can be used as a first codebook. Because the communities overlap, a multi-label coding strategy would be preferable to forcing each article into a single category.